Doc 3 — Report Template

Use this after the visit to turn field data into a portfolio deliverable. Each section lists what goes in it and why an employer cares.

Why this document is the actual deliverable

A heatmap screenshot proves you can run an app. A structured report proves you can scope work, measure methodically, interpret RF behavior, and communicate to a non-technical stakeholder. For junior network, NOC, and wireless roles, the second is what gets you the interview — and it is the thing you can actually talk through for fifteen minutes when someone asks about it.

Section 1 — Executive summary

Half a page, written last, aimed at someone who will read nothing else. Three or four sentences of what you did, followed by your top three findings as plain statements, followed by your top recommendation. No jargon, no dBm figures in this section.

Section 2 — Scope and objectives

- Site, date, times, areas covered and areas explicitly excluded.
- Survey type: passive site survey with limited active spot checks.
- What you were and were not authorized to do — state the passive-only boundary explicitly. This sentence is worth more to a hiring manager than any heatmap.
- Stated objective: characterize existing coverage, identify gaps, identify interference, recommend improvements.

Section 3 — Methodology

This is the section that separates a professional from a hobbyist. Be specific enough that someone could reproduce your work.

Element	What to state
Tool	NetSpot for Android (version), Survey mode, passive
Device	Motorola Moto G Stylus 5G (2024) — Snapdragon 6 Gen 1, Wi-Fi 5 (802.11ac), dual-band, 1×1 single spatial stream, internal antenna
Band coverage	**2.4 GHz and 5 GHz only. No 6 GHz visibility.** State whether the site was believed to have 6E/7 equipment, and that this band was outside the survey's reach
Measurement uncertainty	Consumer uncalibrated chipset; absolute RSSI carries roughly ±3–6 dB uncertainty. Findings rest on relative comparison under a consistent method, not on absolute accuracy
Throughput ceiling	1×1 radio caps observed rates at ~433 Mbps on 80 MHz. Speed test figures describe the client, not the network's capability
Other known limits	Single-client perspective; no spectrum analyzer, so non-WiFi interference is inferred from SNR rather than observed directly; AP positions are trilaterated estimates, cross-checked against physical observation; scan throttling disabled to permit adequate sample rate
Floor plan source	Ground, Second and Third level plans from Architectural Record's 2011 feature on the building — design drawings, not as-builts (the 'Future Aquarium' label is the tell). Credit the source. List every drawing-vs-reality discrepancy you observed
Scale calibration	Reference feature used and its measured length

Element	What to state
Data point density	Count per zone, approximate spacing, and resulting map coverage % — noting that map coverage measures spatial spread, not density
Conditions	Time of day, occupancy, anything that would differ on another day
Measurement discipline	Consistent device height and orientation, stationary during collection

Stating your limitations is a strength, not a weakness. A survey report that claims no limitations tells the reader you don't know what they are.

Section 4 — Environment as found

- SSID inventory (Table 1) — the library's networks and their configuration.
- AP inventory (Table 4) — count, locations, mount types.
- Channel plan as deployed (Table 2) — what channels, what widths, on both bands.
- Neighbouring networks detected and their contribution to the noise/interference picture.
- Building construction notes (Table 6).

Section 5 — Findings

The core of the report. Use this structure for every single finding, numbered:

Field	Content
Finding #	F-01, F-02...
Severity	High / Medium / Low — and say what your severity criteria are
Location	Specific area, referenced to the floor plan
Observation	What you measured. Actual numbers.
Evidence	Heatmap excerpt, screenshot, table reference
Probable cause	Your RF reasoning — distance, attenuation, channel contention, AP placement
Impact	What a patron or staff member actually experiences
Recommendation	Cross-reference to Section 6

Give vertical propagation its own subsection. A three-storey building with an open atrium is not three independent floors — it is one RF volume with a chimney through it. Compare readings at the void edge across all three levels against readings at the same distance from an AP but behind a solid slab. If APs on one floor are serving clients on another, that affects roaming, co-channel planning, and capacity, and almost nobody at junior level thinks to measure it.

Group findings into: **coverage gaps**, **interference and channel planning**, **capacity and density**, **configuration observations**, and **security observations** (signal leakage outside the building, open guest networks, hidden SSIDs that aren't actually hidden, legacy security types).

Section 6 — Recommendations

Ordered by impact-to-effort. Distinguish clearly between the three tiers:

Tier	Examples
No cost — configuration	Channel reassignment to 1/6/11, reduce 2.4 GHz channel width, adjust transmit power, enable band steering, disable low legacy data rates
Low cost — placement	Relocate or re-aim an existing AP, remount from wall to ceiling
Capital — new hardware	Add AP(s) at specified locations; state the coverage gap each one closes

Section 7 — Proposed design (the Planning half)

This is where you finally use NetSpot's Planning mode, at your desk. Build a predictive project on the same floor plan, draw in the walls with the materials you observed, place the existing APs where you actually found them, then add your proposed APs. Present measured-vs-predicted side by side.

Measured problem, modeled solution, on the same floor plan. That pairing is the single most impressive thing you can put in front of a wireless hiring manager, and it costs you one evening.

Section 8 — Appendices

- Full heatmaps: Signal level, SNR, SIR — per zone
- All Inspector screenshots (Networks, Channels 2.4, Channels 5, Comparison)
- Photographs of APs and the space
- Completed field tables, transcribed
- Raw NetSpot project export (.nsp) reference
- Glossary for non-technical readers

Turning this into portfolio and interview material

Deliverables to produce

- [ ] Full PDF report (this template, filled in) — the artifact you send.
- [ ] A one-page summary version — the artifact you hand to a recruiter.
- [ ] A short write-up for your site or LinkedIn: what you set out to learn, what surprised you, what you'd do differently. Process narrative reads far better than a results dump.
- [ ] A copy offered to the library, if they were receptive.

Where this maps to CWNA

Tomorrow's walk touches these CWNA domains directly — note them, because 'I studied it' and 'I did it and here's the report' are different sentences in an interview:

CWNA area	What you'll have practiced
RF fundamentals	dBm, attenuation through real materials, free-space loss over real distances
RF math and measurement	SNR interpretation, reading RSSI against design thresholds
Channel planning	2.4 GHz 1/6/11, 5 GHz UNII bands, DFS, channel width tradeoffs
Interference	CCI vs ACI identified in a live multi-tenant environment
Site survey methodology	Passive survey execution, data point density, calibration, documentation
WLAN design	Coverage vs capacity, cell overlap, AP placement reasoning
Security basics	Signal leakage beyond the building envelope, SSID and security-type observations

Questions you should be able to answer afterward

If you can answer these without notes, the trip did its job:

- Why is –67 dBm the number people design to, rather than –80?
- You measured strong signal but poor SNR in one area. What are the candidate explanations, and how would you tell them apart?
- Two APs are on channel 6 and can hear each other. Describe exactly what happens to throughput and why.
- Why is adjacent-channel interference worse than co-channel interference?
- Your heatmap showed 20% map coverage. Why does that make the whole picture untrustworthy?
- How would your results differ if you ran the same survey at 3pm on a Saturday instead?
- What would you have done differently with professional gear (a spectrum analyzer, a multi-radio survey adapter)?

Is a phone-based survey good enough to showcase?

Yes — for the roles you are targeting, and with conditions. Here is the honest accounting.

What it does prove	What it does not prove
You understand passive survey methodology and can execute it systematically	That you can operate Ekahau or AirMagnet, which is what enterprise WLAN shops actually run
You can read RSSI, SNR, and interference and reason about RF causes	Spectrum analysis — you cannot see non-WiFi interference, only infer it

What it does prove	What it does not prove
You can scope work, get permission, and document to a professional standard	Multi-AP enterprise design at scale, roaming validation, or capacity planning under real load
You understand your instrument's limits and say so unprompted	Anything about 6 GHz, Wi-Fi 6/6E behavior, or multi-stream performance
You finish things and produce a deliverable	Client-facing experience with a paying stakeholder

For a junior network engineer, NOC, wireless, or security analyst role, the left column is more than enough to be a strong talking point. Most candidates at your stage have certifications and nothing they physically did. The gap between 'I'm studying CWNA' and 'here's a survey I ran at a public library, here's the report, here's what I'd fix' is enormous.

How to make it clearly better than a phone screenshot

Ranked by how much they raise the ceiling, cheapest first:

Upgrade	Cost	What it adds
Filter heatmaps by network (include/exclude)	Free, 10 min	Separate library-only, per-SSID, and neighbours-only maps. Looks deliberate rather than dumped.
Two snapshots — quiet period and busy period	One extra visit	Coverage vs capacity on identical geometry. Genuinely sophisticated for a junior portfolio.
Pair the survey with a Planning model	One evening	Measured problem next to modeled solution, same floor plan. The single highest-value addition.
Survey a second, different venue	One more trip	Two sites means you have a repeatable method, not a one-off. A coffee shop or coworking space contrasts nicely with a library.
Write the process narrative, not just results	An hour	What you expected, what surprised you, what you would do differently. Hiring managers read this closely.
Import the .nsp into NetSpot desktop (trial or Home)	Low	Unlocks noise level, AP quantity, PHY mode, and frequency band heatmaps from the data you already collected on the phone. Same walk, considerably more analysis.
Borrow or buy a USB survey adapter + laptop	Real money	Multi-stream, calibrated, 6 GHz capable. Only worth it if you decide to specialize in wireless.

> **The highest-leverage one is the Planning pairing.** It costs an evening at your desk and it is the thing that turns 'I measured a building' into 'I diagnosed a problem and designed a fix.'

Honest framing when you present it

Describe it as what it is: a self-directed practice survey using a consumer smartphone and a mobile survey app, conducted with the venue's permission. Do not dress it up as a professional engagement. Interviewers in this field can tell instantly, and the accurate version is already impressive for someone at your stage — you went out, did the work, and documented it properly. That is more than most candidates bring.